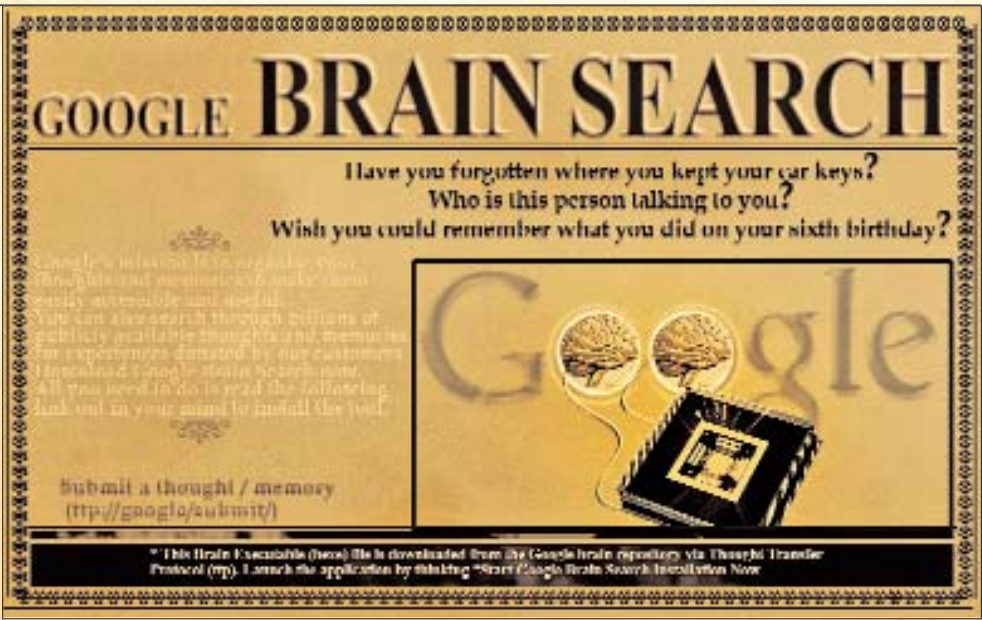


computer? Paranoia is the leveller of the over-enthusiastic, and some believe that mass mind control already exists—visit <http://www.rense.com/general17/imp.htm> and <http://www.parascope.com/articles/0197/mcveigh.htm>, which are indicative.

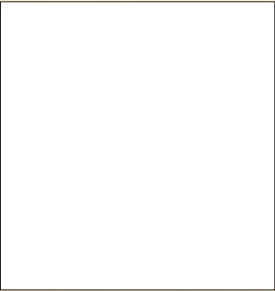
Rat’s Brain Flies Airplane

A brain in a dish! Seriously! Thomas DeMarse, a University of Florida (UF) scientist, grew a living “brain” that could fly a simulated plane. The “brain” was a collection of 25,000 living neurons taken from a rat’s brain and cultured in a glass dish. DeMarse’s experimental “brain” interacts with a flight simulator of an F-22 fighter jet through a desktop computer, and a specially designed plate called a multi-electrode array. DeMarse says, “Over the electrode grid in the dish, we put living cortical neurons from rats, which rapidly begin to re-connect themselves, forming a living neural network—a brain.” The brain and the simulator establish a two-way connection, similar to how neurons receive and interpret signals from each other to control our bodies. To control the simulated aircraft, the neurons first receive information from the computer about flight conditions—for example, whether the plane is flying level or if it is tilted. The neurons then analyse the data and respond by sending signals to the plane’s controls. Those signals alter the flight path and new information is sent to the neurons, creating a feedback system. The “brain” slowly learns to fly the plane, through trial and error. So is this just another neural network displaying learning behaviour? In a sense, yes. But the fact that it is living makes it a milestone. If a rat’s brain cells can pilot a plane, what might a human brain in a dish be capable of?



The Artificial Heart

October 2004, federal regulators approved use of an artificial heart, marking the first time a mechanical



device has been made widely available to replace a failing human organ. In contrast with early hopes of permanently replacing an ailing heart with a man-made pump, the new device substitutes for the bottom half of the heart and can be used only to sustain critically ill patients in the hospital, until they can get a heart transplant. The CardioWest Total Artificial Heart offers hope to Americans who would otherwise die each year without it. “It’s a milestone,” said Robert Jarvik, whose Jarvik-7 heart caused a world buzz when it was implanted in Barney Clark in 1982.

Electronic Eel

Israel: Dr. Amir Karniel, an expert in electrical engineering from the Technion, and a researcher in the area of motor control, has participated in several fascinating experiments. In one of them, he and other researchers separated the brain stem of a lamprey—an eel-like fish—from its body, and kept it alive in a container filled with oxygenated salt water, with electrodes attached to its side. They connected a standard robot with wheels and light sensors to this floating brain. The robot was surrounded by a ring of light bulbs, controlled by the researchers. When a light bulb was flashed, the robot would send a signal to the floating brain, informing it of a flashing light. The brain was able to understand this data, and send back a signal, commanding the robot to move, correctly, towards the light. Now we know that disembodied brains can indeed command robotic appliances to act on their behalf. Perhaps the people who have spent millions on cryogenics (freezing human brains, with the hope of living again

someday) will have the last laugh after all. But living forever, as utopian as it may sound, is not the be-all and end-all of such research. **Wire Your Brain To The Net** Rodney Brooks, a leading roboticist and Professor of Computer Science at MIT, has said, “Direct neural control of complex machines is a long-term US military goal. DARPA has a brain-machine interface program aimed at creating next-generation wireless interfaces between neural systems and, initially, prosthetics and other biomedical devices.” Brooks, who is also Director of the MIT AI Lab, goes on to say that the brain-Internet interface is not as far off as we think. Imagine that - broadband to your brain! (Visit <http://wireless.newsfactor.com/perl/story/22841.html> to see the full article.) For a roundup of some exciting brain implant research, head to http://www.geocities.com/skews_me/implants.html.